

Clemens Thalhammer

PHD STUDENT · APPLIED MATHEMATICS

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Education

ETH Zurich

PHD APPLIED MATHEMATICS

- Supervisor: Habib Ammari

Zurich, Switzerland
since 2025

ETH Zurich

MSC APPLIED MATHEMATICS

- Focus in Mathematical Physics, Graduated with Distinction. Grade 5.78/6

Zurich, Switzerland
2023 - 2025

ETH Zurich

BSc MATHEMATICS

- Grade 5.24/6

Zurich, Switzerland
2020 - 2023

Professional Experience

- 2022, 2023 **Undergraduate Teaching Assistant**, ETH Zurich
2017-2019 **Coordinator**, Kinderuni Medizin, Medical University of Vienna
2013-2017 **Volunteer**, Kinderuni Medizin, Medical University of Vienna

Research Experience

Imperial College London - Dept. of Mathematics

ADVISORS: PROF. HABIB AMMARI, PROF. RICHARD CRASTER, DR. BRYN DAVIES

- MSc Thesis: "Computational Models for Quasiperiodic Differential Equations"

London, United Kingdom
Summer 2024

Teaching Experience

- 2023 **Numerical Methods**, Exam Preparation Course
2023 **Numerical Methods**, Teaching Assistant
2022 **Numerical Methods**, Teaching Assistant

Awards, Fellowships, & Grants

- 2025 **ETH Medal for an outstanding Master Thesis**, ETH Zurich

Presentations

INVITED TALKS

Summer 2026. **A Real-Space Formulation of the Zak Phase via Weyl m-Functions**. Invited talk: MetaMAT, Ajaccio, France.

Spring 2025. **Computational Models for Quasiperiodic ODEs**. Invited talk: Workshop on Metamaterials with Extra Dimensions, London, United Kingdom.

Leadership & Service

LEADERSHIP

VMM (Mid-level Faculty Association)

CO-PRESIDENT

Represented early-career researchers in faculty governance and contributed to policy development and strategic decisions. Acted as a liaison between faculty leadership and members, ensuring transparent communication and engagement.

Zurich, Switzerland
since 2025

SERVICE AND OUTREACH

Kinderuni Medizin, Medical University of Vienna

Vienna, Austria

COORDINATOR

2017-2019

Led the planning and execution of workshops for children, coordinating volunteers, scheduling events, and ensuring smooth program delivery for large audiences.

Publications

Habib Ammari, Silvio Barandun, Yannick De Bruijn, Ping Liu, Clemens Thalhammer. 2025. Spectra and pseudo-spectra of tridiagonal k -Toeplitz matrices and the topological origin of the non-Hermitian skin effect. *Journal of Physics A: Mathematical and Theoretical* 58(20):205201.

Bryn Davies, Clemens Thalhammer. 2025. Convergence of supercell and superspace methods for computing spectra of quasiperiodic operators. *Proceedings of the Royal Society A* 481(2319):20240894.

PREPRINTS

Habib Ammari, Clemens Thalhammer. 2026. A Real-Space Formulation of the Zak Phase via Weyl m -Functions [arXiv:2601.14816](#).

Habib Ammari, Clemens Thalhammer, Alexander Uhlmann. 2025. Uniform Hyperbolicity, Bandgaps and Edge Modes in Aperiodic Systems of Subwavelength Resonators. [arXiv:2509.22417](#).